

Implied Volatility & Skew

Lesson Explainer

The Concept

Implied volatility (IV) is the volatility level that, plugged into an options pricing model, produces the option's actual current market price. Unlike realized volatility (backward-looking), IV is a forward-looking, market-implied forecast embedded in option prices.

IV is rarely flat across strikes. Plotting it typically produces a curve - the volatility skew. For equity index options, the skew is usually downward-sloping: out-of-the-money puts trade at higher IV than calls, reflecting persistent demand for downside protection.

A steepening skew (puts relatively more expensive) signals rising demand for crash protection, often a sign of building fear before the stock has moved much. Reading the skew reveals what the options market is actually worried about.

Illustrative Example: A Downward-Sloping Skew

Implied volatility across five strikes on a hypothetical index option, stock at \$100.

Strike	Type	Moneyness	Implied Volatility
\$85	Put	15% OTM	28%
\$95	Put	5% OTM	21%
\$100	Both	At the Money	18%
\$105	Call	5% OTM	16%
\$115	Call	15% OTM	14%

The deep out-of-the-money put carries nearly double the IV of the equivalent call - reflecting the market pricing in persistent demand for downside crash protection, a pattern seen consistently in equity index options.

5 Things to Know About Implied Volatility

1. IV is forward-looking, realized volatility is backward-looking - they can diverge substantially.
2. IV tends to spike around known events, then drop sharply after - 'volatility crush'.
3. Equity index skew is usually downward-sloping - crash protection demand makes puts pricier.
4. A steepening skew often signals rising fear, even before the stock has actually moved much.
5. High IV doesn't guarantee a big move - it means a wider range of outcomes is priced as likely.

4 Things to Check Before Trading Around IV

- Compare IV to its own history - a level only means something relative to recent IV.
- Watch for volatility crush - buying options before a known event risks IV collapsing after.
- Read the skew, not just one IV number - the shape across strikes carries real information.
- Connect back to vega - vega tells you exactly how much an IV shift moves your position.

Worth knowing:

this explains implied volatility and skew using illustrative numbers - it isn't personalized financial advice, and no reading or level here is a recommendation to trade. Implied volatility is a market-derived estimate, not a guarantee of future price movement.

Quiz — Answer the questions, then check the key on the next page

1. What is implied volatility?

- A. How much a stock has already moved in the past year
- B. The option's strike price divided by the stock price
- C. The volatility level that, plugged into a pricing model, produces the option's current market price
- D. A guaranteed forecast of the stock's future price

2. What shape does equity index volatility skew usually take?

- A. Perfectly flat across every strike
- B. Downward-sloping, with out-of-the-money puts trading at higher IV than calls
- C. Upward-sloping, with calls always more expensive than puts
- D. Random, with no consistent pattern ever observed

3. In the illustrative example, what did the wide gap between put and call IV suggest?

- A. The options were mispriced by mistake
- B. Calls always cost more than puts
- C. The stock was guaranteed to fall
- D. Persistent demand for downside crash protection, a pattern seen consistently in index options

4. What is "volatility crush"?

- A. A sharp drop in implied volatility right after a known event like earnings
- B. A stock crashing more than 50% in one day
- C. An option expiring worthless
- D. A broker closing a position without permission

5. Does high implied volatility guarantee a stock will move a lot?

- A. Yes, IV is a guaranteed forecast
- B. No, IV has no relationship to expected movement at all
- C. No, it means a wider range of outcomes is being priced as more likely, not a guarantee of one
- D. Yes, but only for put options

Answer Key

1. What is implied volatility?

Correct answer: C. The volatility level that, plugged into a pricing model, produces the option's current market price

Implied volatility is a forward-looking, market-implied estimate embedded in the option's current price.

2. What shape does equity index volatility skew usually take?

Correct answer: B. Downward-sloping, with out-of-the-money puts trading at higher IV than calls

Equity index skew is usually downward-sloping, reflecting persistent demand for crash protection.

3. In the illustrative example, what did the wide gap between put and call IV suggest?

Correct answer: D. Persistent demand for downside crash protection, a pattern seen consistently in index options

The 28% put IV versus 14% call IV reflects the market's ongoing willingness to pay up for downside protection.

4. What is "volatility crush"?

Correct answer: A. A sharp drop in implied volatility right after a known event like earnings

Volatility crush refers to IV spiking ahead of an event, then dropping sharply once uncertainty resolves.

5. Does high implied volatility guarantee a stock will move a lot?

Correct answer: C. No, it means a wider range of outcomes is being priced as more likely, not a guarantee of one

High IV reflects a wider expected range of outcomes being priced in, not a certainty of any specific move.